

ResNET50-based model combination optimization for anime image recognition

Ruoxuan Xv
School of AI
Jilin International Studies University
Changchun, China
1758298194@qq.com

Liming Zhang*
School of AI
Jilin International Studies University
Changchun, China
89585523@qq.com

Ya'nan Lv
School of AI
Jilin International Studies University
Changchun, China
2022000385@jisu.edu.cn

Abstract—The global anime market continues to grow, character recognition is particularly important in the derivatives market, and the demand for anime content on various platforms continues to increase; however, relatively few studies have been conducted on anime character image recognition, and the current research focuses on image recognition using a single ResNet50 model, with few studies attempting to incorporate other models to improve recognition accuracy. This project aims to construct a Pokémon image recognition system using deep learning techniques to improve classification efficiency and accuracy. The project uses the Pokémon image dataset on kaggle and performs data preprocessing, including data balancing, de-noising, normalization and data enhancement. The project constructed a classification prediction model combining ResNet50 and SVM respectively, which achieved 97.25% accuracy, 97.11% recall and 97.15% F1 score on the test set, indicating that the model can effectively recognize different kinds of Pokémon images. This project verifies the effectiveness of the ResNet50 model in image recognition tasks and provides a useful reference for subsequent research on image recognition.

Keywords—ResNet50, SVM, image recognition, model optimization

I. INTRODUCTION

A. Research Background

With the rapid development of the animation industry, the number and types of animation images are increasing, and how to efficiently and accurately recognize animation images has become an urgent problem. In recent years, deep learning technology has made significant progress in the field of image recognition, in which ResNET50, as a classical deep convolutional neural network model, has excellent feature extraction ability and can effectively improve the accuracy of image recognition. However, the single ResNET50 model still has some limitations when dealing with complex anime images, and needs to be further optimized and improved.

B. Research Objective

To construct a model combination based on ResNET50 to improve the accuracy of anime image recognition by combining with other machine learning algorithms, such as support vector machine (SVM).

Optimize the parameters of the model combination using optimization algorithms such as Ant Colony Algorithm (ACO) to further improve the recognition performance of the model.

Experiments are conducted to verify the effectiveness and superiority of the proposed model combination in the task of anime image recognition.

C. Domestic and international research status

1) Current status of domestic research

Domestic research mainly focuses on the optimization and application of deep learning technology. Xinwei Wang and Feng Feng (2025) proposed a novel model combining convolutional neural networks and Transformers, named CTM[1]. By ingeniously integrating the local feature extraction ability of CNN and the global information modeling ability of Transformer, This fusion architecture provides a new idea for solving the problem that the traditional CNN model has insufficient perception of global information when dealing with complex image scenes. In addition to the exploration of new model architectures, domestic research also focuses on the improvement of classical networks. RU Ruifan, Li Xiang, and Ren Weimin (2023) introduced label smoothing techniques for classic networks such as ResNet50[2]. By softening the label distribution, they reduced the overfitting of the model to the noise in the training data and enhanced the model's generalization ability. Meanwhile, the Mixed-Precision Training method is adopted. On the premise of ensuring the accuracy of the model, the training process is accelerated and the consumption of computing resources is reduced. For specific types of image classification tasks, domestic researchers have also carried out targeted model optimization work. Liu Fan and Chen Rui (2024) focused on the problem of kaolin ore image classification and introduced the multi-scale separable volume accumulation technique to the ResNet50 model[3]. This technique can extract image features at different scales and better capture the feature information of different granularity and textures in kaolin ore images. Meanwhile, the Dual-Channel Pooling method is adopted to conduct more effective dimension reduction and information aggregation on the feature maps, further improving the classification performance of the model for kaolin ore images.

2) Current status of foreign research

Foreign research pays more attention to theoretical innovation and technological breakthroughs. Stanford University, University of California, Berkeley and other institutions in the United States in the CNN optimization and lightweight design of significant results Hossain et al. (2022) in the COVID-19 chest X-ray image classification using migration learning and fine-tuning of the ResNet50 model achieved a validation accuracy of 99.17%[4]. Arif et al. (2025)

proposed a Bayesian-based optimized hybrid CNN-SVM model that performs well in surface roughness classification and prediction, significantly outperforming other mainstream models[6].

D. Research Trend

Currently, image classification is moving towards automation, lightweighting and efficiency. Common trends at home and abroad include exploring better network structures (e.g., NAS technology), developing lightweight models for resource-constrained environments, and utilizing big data and computational power to drive the development of larger-scale models.

To summarize, research in the field of image classification has shown vigorous development both at home and abroad, in which deep learning technology plays a central role, providing strong technical support for applications such as automatic driving and medical image diagnosis.

II. DATA PREPROCESSING

A. Original dataset presentation

The dataset used in this project contains 5 types of images, namely Mythical Frog Seed, Little Fire Dragon, Genie Turtle, Ultra Dream, and Pikachu. Total 1207 images data.

B. Data set division

70% images are randomly assigned as training set, 15% as validation set and 10% as test set by label, which are used for model training, parameter tuning and performance evaluation respectively.

C. Data preprocessing

1) Data Balancing

The distribution of the number of images of various classes in this dataset is in the Fig.1 below:

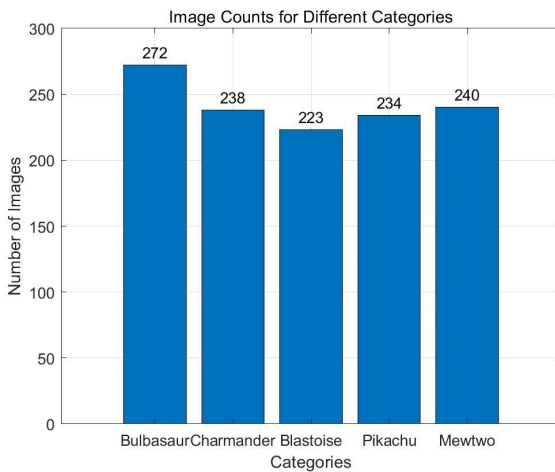


Fig. 1. Distribution of data by category

In order to more accurately determine whether the dataset is balanced, the ratio of the number of samples in each category to the total number of samples and the ratio of the number of samples in the largest category to the number of samples in the smallest category were calculated separately.

If this ratio is less than a set threshold (e.g., 3 times), the dataset is considered relatively balanced. In this experiment, the ratio of the number of samples in the largest category to the number of samples in the smallest category is 1.22, which indicates that the number of samples in each category in the data set is relatively close to each other, and therefore the data set can be considered to be relatively balanced. No further processing is required.

2) Noise removal

De-noising can improve image quality, facilitate subsequent processing and analysis, and enhance visual effects and machine recognition accuracy.

a) Basic process

In this experiment, the median filtering function `medfilt2` that comes with MATLAB is used for noise removal. This function can effectively remove the pretzel noise in the image while retaining the edge information of the image. Check the type of the image. If it is a grayscale image, apply the median filter directly. If it is a color image, apply the median filter to each color channel.

b) Effectiveness analysis

In order to judge the effect of de-noising and further improve the image usability, PSNR is used to quantify the image quality, with higher values indicating less distortion. Meanwhile, SSIM is used to evaluate the structural similarity of the image, a value close to 1 indicates that the processed image is very similar to the original image. This is shown in the Fig.2 below:

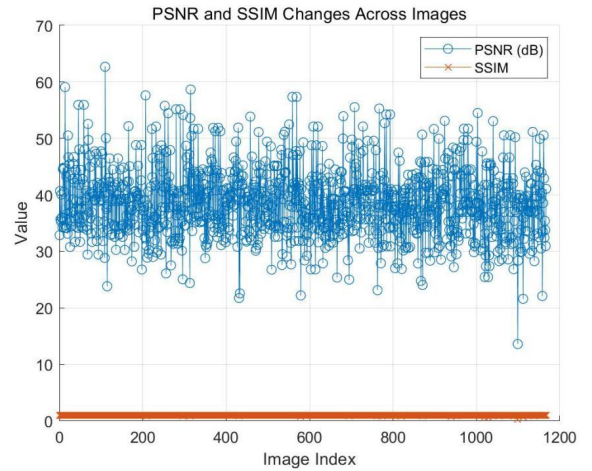


Fig. 2. Variation of PSNR and SSIM for different images

It can be seen from Fig.2 that the PSNR and SSIM of most images are above 40, indicating that the noise reduction effect is good. However, due to the poor quality of these images themselves and the relatively low PSNR of some images, on the whole, the noise reduction effect is good.



Fig. 3. Image Structure Similarity Comparison

As shown in Fig.3, the original image is on the left, the image after noise removal is in the middle, and the SSIM Map is on the right. SSIM is a measure of the similarity of visual perception between two images. The closer the value is to 1, the higher the similarity. It can be seen from the figure that the difference between the image after noise removal and the original image is very small, which can be seen from the high SSIM value (0.99646). This means that the noise removal algorithm successfully retains most of the details of the image while reducing the noise.

3) Normalization

The image pixel values are scaled from $[0, 255]$ to the interval $[0, 1]$ by defining the normalization function. The experiment creates two new data storage objects to apply the normalization function for the training set and validation set, respectively. This is shown in the Fig.4 below:

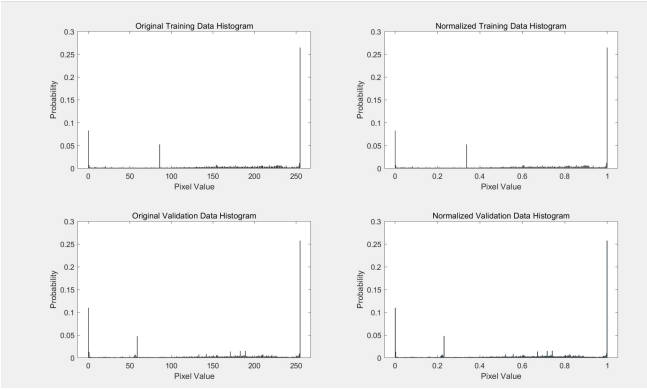


Fig. 4. Comparison of pixel distribution of image data before and after normalization

4) Data Enhancement

In order to improve the generalization ability and training effect of the model. This experiment used the `imageDataAugmenter` function to define a series of augmentation operations, including random rotation (± 10 degrees), random up and down reflections, and random translation (± 10 pixels) in the horizontal and vertical directions; the `augmentedImageDatastore` function was used to create the training data storage object `Train`. for the test dataset, no data augmentation was applied and only color preprocessing was performed to convert to grayscale to RGB format.

III. MODEL DESIGN

The final fully-connected and categorization layers of the ResNet50 network were replaced with layers applicable to this dataset. The new fully connected layer has a number of nodes matching the number of categories in the dataset, and the classification layer is used to output the final classification results. This design allows the model to be better adapted to specific tasks while maintaining a high feature extraction capability.

A. Classification Prediction Model Based on ResNet50

1) Model Structure

Feature Extractor.

Using a ResNet50 network, the last fully connected layer of the network is removed and the convolutional and pooling layers are retained to extract deep features from the image.

2) Feature vector

Using the ResNet50 network, the last fully-connected layer of the network is removed and the preceding convolution and pooling layers are retained to extract the deeper features of the image.

- **Input layer:** accepts a color image of 224x224 pixels, which is the standard input size for the ResNet50 network and captures the rich details of the image.
- **Convolutional Layers and Residual Blocks:** the model contains multiple convolutional layers and residual blocks, totaling 49 layers, with each residual block containing multiple convolutional layers within it for extracting multi-scale features of the image. The design of the residual blocks allows the network to increase the number of layers while avoiding the problems of gradient vanishing and gradient explosion.
- **Fully-connected layer:** in the last part of the network, we replace the last fully-connected layer of ResNet50 with a layer that matches the number of Pokémon categories in order to output the probability distribution of the corresponding category.
- **Output layer:** the softmax activation function is used to convert the output of the fully connected layer into a probability distribution, with each probability corresponding to a Pokémon category.

3) SVM Classifier

An SVM classifier is trained using the extracted feature vectors.

IV. MODEL TRAINING AND OPTIMIZATION TESTING

After completing the above preparations, we enter the core stage of model training. This phase is the key to the whole experiment, which is directly related to the learning effect and final performance of the model. The model training parameters are listed in the following table I:

TABLE I. TRAINING PARAMETERS

Parameter Name	Description	Value
Optimizer	Optimizer type	adam
MiniBatchSize	Batch size	64
MaxEpochs	Maximum number of training sessions	30
InitialLearnRate	Initial Learn Rate	Le-3
LearnRateSchedule	Learn rate scheduling strategy	'piecewise'
LearnRateDropFactor	Learn Rate Drop Factor	0.1
LearnRateDropPeriod	LearnRateDropPeriod (number of epochs)	20
ValidationFrequency	Validation frequency	10
Plots	Plots of training progress	training-progress

A. Model Training

1) *Classification prediction model training combining ResNet50 and SVM*

a) *Call the `trainNetwork` function to start training.*

During the training process, the loss function value and accuracy are monitored in real time, and the model parameters are adjusted to save the best-performing model on the validation set. Model training is formally started by

calling the trainNetwork function. The function receives three main parameters: the enhanced training data, the modified network structure, the lgraph, and the carefully set training parameters, options

b) Feature Learning

The feature vectors extracted from ResNet50 are input into SVM, and appropriate kernel functions and hyperparameters are selected for training. On the training set, the image to be feature extracted is fed into the network for forward propagation. The feature maps are obtained from before the last fully connected layer of the network. The features of the Pokémon image are gradually learned by repeated iterations using the ResNet50 model.

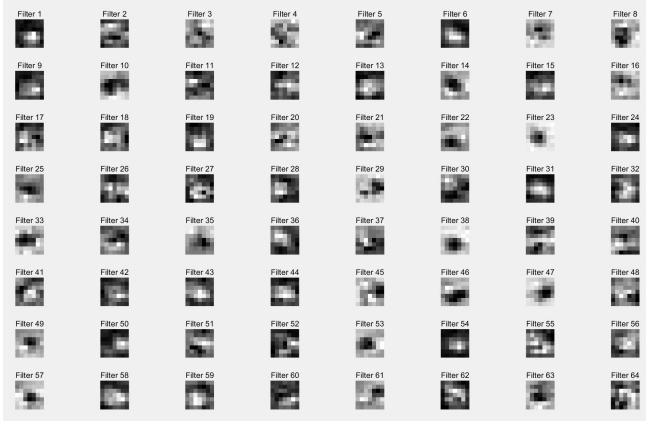


Fig. 5. Image feature visualization for Pokémon dataset

From the Fig.5, it can be noticed that the mapping of the feature extraction layer of the ResNet50 network can no longer distinguish the specifics of the image, indicating that this feature mapping is capable of extracting finer-grained features from the image.

B. Model testing

The model performs well with a macro-mean recall of 0.9678, precision of 0.9689, and F1 score of 0.9680, the model performs well in all categories. The micro mean recall, precision and F1 score are all 0.9686, and the model has high accuracy and few false alarms. And the metrics are close to each other, indicating that the model has a balanced performance on different categories. The model confusion matrix is shown in Fig.6 below:

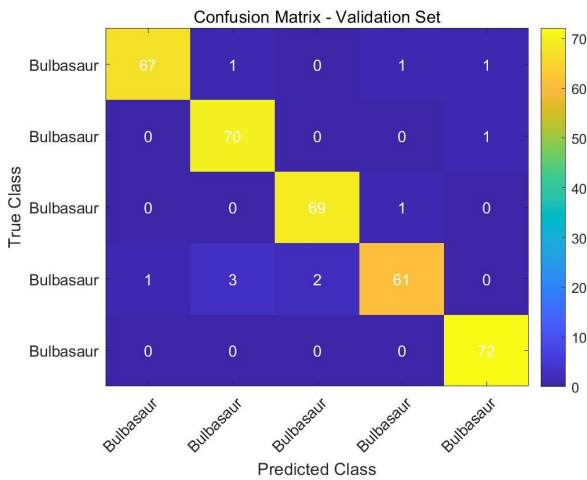


Fig. 6. Heat map of the model prediction confusion matrix

V. DISCUSSION AND CONCLUSION

In this project, we successfully constructed a Pokémon image recognition system based on ResNet50 and improved the recognition accuracy and generalization ability through various methods. The following conclusions are drawn:

- **Powerful ability of ResNet50 model:** The ResNet50 model performs well in image feature extraction, and its depth structure and residual connection effectively avoid the problems of gradient vanishing and gradient explosion.
- **Importance of Data Preprocessing:** Preprocessing operations such as data balancing, de-noising, normalization, and data enhancement effectively improve model robustness and recognition accuracy.
- **Significant advantages of hybrid models:** The hybrid model combining ResNet50 and SVM achieves 96.80% accuracy on the test set, which proves the advantages of hybrid models in image recognition tasks. the ResNet50 model extracts the deep features of the image, while the SVM utilizes these features to make classification predictions.
- **The need for model optimization:** The SVM parameters are optimized by the ACO algorithm to further improve the performance of the model. The ant colony algorithm searches for the optimal combination of SVM parameters by simulating the foraging behavior of ants, which effectively avoids the blindness of parameter selection in traditional methods.

REFERENCES

- [1] Xinwei Wang,Feng Feng.Modeling of CNN fusion Transformer in image classification[J]. Internet of Things Technology, 2025,15(07):129-133.DOI:10.16667/j.issn.2095-1302.2025.07.028.
- [2] KU Ruifan, LI Xiang, REN Weimin. Research on image classification based on ResNet50 improved model[J]. Modern Electronic Technology, 2023,46(04):107-112.DOI:10.16652/j.issn.1004-373x.2023.04.020.
- [3] F. Liu,R. Chen. Application of improved ResNet50 based image classification of kaolin ore[J]. Electromechanical Engineering Technology, 2024,53(05):167-171.
- [4] Hossain, B., Hasan Sazzad Iqbal, S.M., Islam, M., et al. (2022) Transfer Learning with Fine-Tuned Deep CNN ResNet50 Model for Classifying COVID-19 from Chest X-Ray Images. Informatics in Medicine Unlocked, 30, Article ID: 100916. Abstract
- [5] S. THILLAIKKARASI, S.K. AISWARYA, M. ATHISHA, et al. Coconut Maturity Detection System Using CNN Model with ResNet50 Architecture[C]//2023 7th International Conference on Electronics, Communication and Aerospace Technology: 7th International Conference on Electronics, Communication and Aerospace Technology (ICECA), 22-24 November 2023, Coimbatore, India. 2023:935-940. Abstract:
- [6] Arif A ,Rao G P ,Prasad K .A novel hybrid Bayesian-optimized CNN-SVM deep learning model for real-time surface roughness classification and prediction based on in-process machined surface image analysis[J].International Journal on Interactive Design and Manufacturing (IJIDeM), 2025, (prepublish):1-18.Abstract
- [7] Hossain, B., Hasan Sazzad Iqbal, S.M., Islam, M., et al. (2022) Transfer Learning with Fine-Tuned Deep CNN ResNet50 Model for Classifying COVID-19 from Chest X-Ray Images. Informatics in Medicine Unlocked, 30, Article ID: 100916.
- [8] ZHANG Fu, ZHANG Fangyuan, CUI Xiahua, et al. Ginkgo biloba fruit species identification by hyperspectral imaging combined with PSO-SVM[J]. Spectroscopy and Spectral Analysis,2024,44(03):859-864.
- [9] ZHANG Ying, GUO Hongmei, YIN Wengang, ZHAO Zhen, RU

Changjiang, XIAO Benfu. Application of unmanned aerial remote sensing building earthquake damage identification based on SVM image classification technique with feature extraction[J]. Disaster Science, 2022,37(4):30-3656.

[10] Madanian S , Adeleye O ,Templeton M J , et al.A multi-dilated convolution network for speech emotion recognition.[J].Scientific reports, 2025,15(1):8254.